



Gennoor Tech
TRAIN. INNOVATE. BUILD.

5-DAY HANDS-ON BOOTCAMP · 40 HOURS

Deep Learning & Computer Vision

Neural networks, CNNs, and computer vision in practice

5 40 15+ 5+

DAYS HOURS HANDS-ON LABS PROJECTS

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Course Overview

This 5-day hands-on bootcamp covers everything from foundational concepts to advanced techniques, combining lectures, live demos, hands-on labs, and collaborative workshops for maximum learning impact.

Prerequisites: Python (intermediate), basic ML knowledge, linear algebra fundamentals.

Who Is This For?



ML Engineers



Researchers



CV Engineers



L&D Teams
Internal trainers

Learning Outcomes

- 1 **Build neural networks** — from scratch with NumPy, then TensorFlow/PyTorch
- 2 **Master CNNs** — ResNet, EfficientNet for classification and detection
- 3 **Apply RNNs, LSTMs,** — and Transformer architectures
- 4 **Build production vision** — pipelines with YOLO, OCR, and Azure AI Vision
- 5 **Deploy deep learning** — models with ONNX and Docker

5-Day Curriculum

Interactive format with live demos, hands-on labs, and collaborative workshops. Schedule and session order are fully customizable based on your organization's needs and time zones.

01

Neural Network Foundations

HANDS-ON LABS + WORKSHOP

- SESSION A
- Neural Network Foundations
- Neurons, activation functions (ReLU, sigmoid, softmax)
 - Backpropagation and gradient descent (SGD, Adam)
 - Regularization: dropout, batch normalization, L1/L2

- SESSION B
- Hands-On & Practice
- TensorFlow 2.x: Sequential and Functional APIs
 - Lab: Build MNIST classifier in TensorFlow and PyTorch

02

Convolutional Neural Networks

DEMOS + LABS + DISCUSSION

- SESSION A
- Convolutional Neural Networks
- Convolution, pooling, stride, padding — visual intuition
 - Classic architectures: LeNet, VGG, ResNet, EfficientNet
 - Transfer learning and fine-tuning pretrained models

- SESSION B
- Hands-On & Practice
- Data augmentation and Grad-CAM visualization
 - Lab: Medical image classification & defect detection

03

Sequence Models & Transformers

LECTURE + HANDS-ON LABS

SESSION A

Sequence Models

- RNNs, LSTMs, GRUs — architecture and training
- Bidirectional and stacked RNN patterns
- Self-attention, multi-head attention, positional encoding

SESSION B

Transformers

- BERT, GPT overview, Vision Transformers (ViT)
- Lab: Time-series forecasting & BERT text classification

04

Object Detection, Segmentation & OCR

HANDS-ON LABS + WORKSHOP

SESSION A

Object Detection, Segmentation

- YOLO (v8/v11), SSD, Faster R-CNN architectures
- Instance segmentation (Mask R-CNN), semantic segmentation (U-Net)
- OpenCV: image processing, contours, transformations

SESSION B

OCR

- OCR: Tesseract, EasyOCR, Azure Document Intelligence
- Lab: Real-time detection system & invoice extraction

05

Generative Vision, Deployment & Capstone

DEMOS + LABS + DISCUSSION

SESSION A

Generative Vision, Deployment

- Autoencoders, VAEs for anomaly detection
- GANs: DCGAN, diffusion models overview
- ONNX export, TensorRT optimization, quantization

SESSION B

Capstone

- Serving with FastAPI + Docker, edge deployment
- Capstone: End-to-end vision project

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest AI developments.

Tools & Platforms Covered

All tools demonstrated and practiced hands-on during the bootcamp

- Docker
- FastAPI
- PyTorch
- TensorFlow

What You Take Home

 **6 DL Projects (complete notebooks)**
Included with course completion

 **Architecture Reference (CNN, RNN, Transformer guides)**
Included with course completion

 **Vision Pipeline Kit (YOLO + OCR templates)**
Included with course completion

 **Model Zoo Cheat Sheet (accuracy/speed tradeoffs)**
Included with course completion

Certification of Completion

**Gennoor Tech — Certificate of Completion**
Upon completing all 5 days (40 hours), each participant receives a **Gennoor Tech Certificate of Completion** in "Deep Learning & Computer Vision". Digitally signed by the Founder & Director. Validates proficiency in all course topics and hands-on competency.

Recommended Next Certifications

AI-103
Azure AI Solutions

DP-100
Azure Data Scientist

AI-901
Azure AI Fundamentals (New 2026)

Designed & Delivered By

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Senior AI Consultant and **Microsoft Certified Trainer (MCT)** with 14+ years of **16+ active Microsoft certifications** experience. Delivered AI and technology programs for Microsoft, EY, IBM, Boeing, including AB-100 (Agentic AI), AI-103, AZ-400, Saudi Aramco, HDFC Bank, and 80+ organizations across 6+ countries including the AI-300, PL-400, and GitHub certifications. 10-day AI Leadership program for MCIT Saudi Arabia.
Builder of open-source MCP servers and Copilot Studio integrations. M.E. Mechanical Engineering, Mumbai University.

Delivery Options ■ **On-Site** — customized at your office | 🖥️ **Virtual Live** — interactive via Teams/Zoom | 🌐 **International** — GCC, Africa, APAC, UK



Ready to Start Your Deep Learning & Computer Journey?

Contact us to discuss a customized program
for your organization or register for the next open batch.

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